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- Search skills

Shodan
↳ scans the internet, searching for networking equipment, traffic cameras which are in the public network

- VirusTotal
↳ coalets result from over 70 antivirus engines and website scanners
↳ tells what is safe and not safe

- CVE (common vulnerabilites and exposure)
↳ each vulnerability is assigned a unique ID in form CVE-YEAR-NO.

vulnerabilies are given a score (CVSS) based on
→ Impact
→ complexity
→ Availability

To know all Linux ~~co~~ CLI, use Linux Manual page, can be used in terminal only.

man nc

- GitHub
~~greate~~ great resource to stay updated on latest threats & vulnerabilities
Researches publish POC's codes, exploitation tools & technical reports.

LINUS FUNDAMENTAL

PT - 1

- echo
  - output any text we provide

- who ami
  - used befor ; used to know the current user being used

- cat
  - con catenat
  - reads information in a file

we can also used cat as

cat /home/ubuntu/Documents/todo.txt

~~to see~~

- Find
  - To find file(or) data in any part of computer
  - Find -name <filename>

to find only a specific extension use *

∴ Find -name (*).txt

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- grep
  - helps to search specific content of files for specific values

To check all the files in the current directory, use $\boxed{\text{Grep -R}}$

file ke andhar ka ek specific content nikalna

→ grep "THM" home/tryhackme/access.log

- &
  - opertor allows you to run commands on backgrond of terminal

- &&
  - allows to combine multiple commands together in one line
  - only works if prior command was successful

- >
  - opertor is a redirector

Eg if we want a file named "welcome" with message "hey" then use

echo hey> welcome

- >>
  - this avoids possible overwritting in a file

Q According to prev command, if we also want to add tryhackme then use

echo tryhackme >> welcome

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PT-2

- SSH
  - secure shell
  - allows us to remotely execute commands on another device remotely
  - Any data sent b/w them is encrypted

ssh tryhackme@10.80.184.24

For help related ls command use ls --help
For manual use ~~ls man~~ man ls

- touch
  - Create file

- mkdir
  - Create folder

- cp
  - copy a file or folder

- mv
  - move a file or folder (mostly used to rename)

- rm → To remove directory use rm -R
  - remove file (or) folder

- file
  - Determine type of file

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to copy

cp note1 note2
(content is now copied to note2 also)

to mv (rename)

mv note1 note3
(~~co~~ note1 is now renamed as note3)

- **Permission (101)**

[su]
→ helps to switch user

Check THM

Understanding permissions

			numeric values
first 3 → owner		r → read	(4)
next 3 → group		w → write	(2)
last 3 → other		e → execute	(1)

- **Common directory**

[/etc]
→ most important root directory of system
→ location to store system file that are used by OS

_/_/_

passwd ⇒ special for Linux as they show how system stores
shadow the password for each user in encrypted format

sudoers ⇒ contains list of users & grp ~~on~~ who have access to run $\boxed{\text{sudo}}$ command

$\boxed{\text{/var}}$
↳ stores data which is frequently accessed by services ~~&~~ (or) applications running on system & also used to visit logs.

$\boxed{\text{/root}}$
↳ /root is the home directory for the main admin (or) root user

$\boxed{\text{/tmp}}$ (Useful for pentesting to store enumeration scripts)
↳ unique root directory
↳ volatile and used to store data that is only needed to be accessed once (or) twice

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PT-3

- Terminal Text editor

$\boxed{\text{nano}}$

↳ helps to create ~~text~~ text editor

$\boxed{\text{nano <filename>}}$

can do's ⇒ search for text
copying and pasting
Jumping to ~~line~~ line number
Finding out what line no

$\boxed{\text{VIM}}$

↳ much more advanced than nano

if want more detail the join VIM room in THM

can do's ⇒ can modify keyboard shortcuts
syntax highlighting
works on all terminals

- Useful utilities

$\boxed{\text{wget}}$

↳ allows to download file from web via HTTP, we just need to provide the address of resource

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- SCP
  - secure copy
  - allows to transfer file b/w 2 computers using SSH

scp important.txt ubuntu@192.168.1.30:/home/ubuntu/transferred.txt

important.txt → name of file in remote system
ubuntu → username
192.168.1.30 → IP address of remote
transferred.txt → name that we wish to store in our system

(or)

scp ubuntu@192.168.1.30:/home/ubuntu/document.txt notes.txt

- Serving files from your host - WEB
  - provides lightweight easy-to-use module called HTTP server

FOR PRACTICAL EXP
MUST VISIT THM

- Process 101
  - programs running on your device
  - managed by the kernel
  - each process has an ID alloted to it, known as PID

$\boxed{\text{ps}}$
- shows all the process running from the user end

$\boxed{\text{ps aux}}$
- To see proccess run by other users

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top
- gives real time statistics about the processes running
- auto refreshes every 10 sec (or) use the arrow keys to browse various rows

kill
- used to terminate process by associating its PID
  eg:- kill 1136

SIGTERM :-
SIGKILL :-
SIGSTOP :-

- How processes start :-
  - process with an ID of 0 is a process that is started when system boots
  - systemd is a way of managing a user's process and sits b/w the OS & user

systemctl
- allows us to interact with the systemd

eg:- systemctl start apache2
systemctl stop
enable → Starts the service while boot up of system
disable
status

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- Background & foregrounding

echo "Hi THM" &

↳ given us the ID of the process as it is now running in the background

$\boxed{\text{Ctrl +Z}}$ → used to pause the process

$\boxed{\text{fg}}$

↳ used to bring previously backgrounded process to foreground

- Automation

crontabs

↳ started during boot, responsible for facilitating & managing cron jobs

$\boxed{\text{Cron}}$

↳ requires 6 specific values

→ MIN :- what minute to execute at
→ HOUR :- " hour " " "
→ DOM :- " day of month to execute at
→ MON :- " month of year " " "
→ DOW :- " day of week " " "
→ CMD :- The actual command that is executed

jo value nhi daalni usse * mai krdo

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crontab edit krne ke liye use $\boxed{\text{crontab -e}}$
To open crontab use $\boxed{\text{crontab -l}}$

- Package Management

$\boxed{\text{apt}}$
→ if developers wish to submit softwares, they will submit to apt repository

$\boxed{\text{add-apt-repository}}$

we can use $\boxed{\text{apt}}$ to install softwares
also can use $\boxed{\text{dpkg}}$

when downloading, the integrity is guaranteed by GPG (Gnu privacy guard) keys

THM labs do not have access for internet
∴ not done

- Managing logs

OS has improved the way it stores log, which is known as rotating

2 types of log files
→ Access logs
→ error "

Windows Fundamentals

PT-1

- File system

NTFS ⇒ new technology file system
FAT 16/32 ⇒ File allocation table
HPFS ⇒ high performance file system

→ Can automatically repair folder/file on disk using info stored in log file
→ supports file larger than 4GB
→ set specific permission on folder & file
→ Folder & file compression
→ Encryption (EFS)

Permission for NTFS is
→ Full control
→ Modify
→ List folder content
→ Read & Execute
→ Write
→ Read

→ Another feature of NTFS is ADS (Alternate data stream)
→ malware writers use ADS to hide data

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C:\Windows ⇒ place of windows OS

System environment variable ⇒ %windir%

↳ Stores info abt the OS environment includes info such as OS path, no of processor used by OS & location of temporary files

System 32 ⇒ MOST CRITICAL FILE TO RUN OS

- Users & Groups

Adminstrator ⇒ add user, delete user, modify groups
Standard user ⇒ can only make change to folder/file & cannot perform system level changes

also can access through "lusrmgr.msc"

large majority of home users are local adminstrator which can increase the risk of system compromise

UAC (user account control)
when a user with account type of adminstrator logs into a system, the current session doesn't run with elevated permission, if higher-level priveleges needs to execute, the user will be prompted to confirm the operation

The local admin requires UAC but built in admin doesn't need UAC for majority programs

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→ Control panel helps access more complex settings

→ Task Manager provides information about the applications & process currently running on system (ctrl + shift + esc)

PT-2

- System configuration

MS config is for advanced troubleshootings and main purpose is to diagnose startup issues

→ General ⇒ type of startup to load upon boot
→ Boot ⇒ defines various boot options for OS
→ Services ⇒ lists all the services configured for the system regardless of their state (running (or) stopped)

→ Startup ⇒ microsoft advises to use task manger to enable/disable Startup items
NOTE = msconfig is not a sterhup management program

to check startup items, use win + R - then shell:startup

→ Tools ⇒ many tools/utilites we can run to configure the OS

For advanced configuration setting ⇒ search (view advance system setting) windows uses page file as an extra virtual memory space when physical RAM becomes full, it prevents slowdown & crashes

- → In advanced ~~setting~~ system config
  go to Advanced Tab

The startup & recovery can help access crash dump file to check the error behind the crash

Windows has many type of crash dump
→ Auto memory dump
→ kernel " "
→ Small " " (256 KB)
→ Complete " "

UAC setting can be changed (or) totally turned off by searching user account control setting (or)

win + R → useraccountcontrolsettings.exe

- **Computer management**

using compmgmt ⇒ ~~Primary~~ System tools
⇒ Storage
⇒ Services and applications

↓
msc
file type

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→ Task Scheduler :- Automatically carries out common task at specific time

→ Event viewer :- • allows us to view events that have occured on the computer
• can be seen as audit trail that can be used to understand the activity

To see types of events, go to THM

→ Shared folders :- all the folders which shared and can be connected to
in sessions we see the list of users who are currently connected to the shares.

→ Local user & grps :- (All done in PT-1)

→ Performance :- can be accessed by typing "Perfmon"
used to view performance data either in real time (or) log file

→ Device Manager :- helps to view and configure hardware

→ Disk Management :- set up new drive
extend a partition
shrink a "
Assign (or) change drive letter (E!)